

Deep Learning Architectures for RSSI-Based Passenger Movement Classification in Public Transport

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Abstract—Accurate, non-intrusive monitoring of passenger flows is essential for intelligent public transport systems. Prior work demonstrated that temporal sequences of Wi-Fi Received Signal Strength Indicator (RSSI) measurements can classify passenger movements using classical machine learning, achieving a Matthews Correlation Coefficient (MCC) of 0.756 on combined datasets. This paper extends that line of research by systematically evaluating deep learning architectures—including recurrent, convolutional, transformer, state-space, and mixture-of-experts models—for the same classification task across four dataset variants. We introduce 11 novel architectures, a class-conditioned data augmentation strategy expanding 2,253 samples to 10,115, and comprehensive hyperparameter optimization with 50 Optuna trials per model. Our best ensemble model, a two-stage deep stacking architecture, achieves MCC of 0.859 and accuracy of 89.4% on combined data, a 13.6% relative improvement over classical baselines. We further analyze the impact of multi-device interference, per-class confusion patterns, and the effectiveness of metadata-aware architectures in distinguishing true movement from environmental noise.

Index Terms—Passenger counting, RSSI fingerprinting, Wi-Fi sensing, public transport, deep learning, state space models, mixture of experts, hyperparameter optimization

I. INTRODUCTION

Public transport systems serve over 75% of EU urban citizens, yet inefficiencies contribute to 24% of greenhouse gas emissions. Accurate passenger flow knowledge, covering boarding, alighting, and occupancy per stop, is fundamental to network optimization and dynamic scheduling.

Traditional Automatic Passenger Counting (APC) relies on cameras, infrared sensors, or manual counting. Each faces tradeoffs in cost, privacy, and deployment complexity. Wi-Fi sensing offers a privacy-preserving alternative. By analyzing Received Signal Strength Indicator (RSSI) patterns from passengers' devices, operators can infer movement without active user participation or additional hardware beyond existing access points.

RSSI measurements are inherently noisy, affected by multipath propagation, device heterogeneity, and interference from concurrent transmissions. In prior work [1], we introduced a controlled dataset simulating four passenger movements (inside vehicle AA, boarding BA, alighting AB, at stop BB). Classical machine learning achieved MCC 0.756 on combined data using Gaussian Processes. This paper investigates whether modern deep learning architectures can better capture RSSI temporal dynamics, particularly under noisy multi-device conditions.

Our contributions are as follows. First, we systematically evaluate 11+ DL architectures across four dataset variants with increasing interference complexity, including state-space models (Mamba-3), mixture-of-experts, and two-stage stacking ensembles. Second, we propose a class-conditioned data augmentation strategy that expands 2,253 to 10,115 samples through targeted Gaussian noise injection and intra-class Mixup. Third, we conduct comprehensive hyperparameter optimization via 50 Optuna trials per model spanning 19 search spaces. Fourth, we present a detailed interference analysis quantifying cross-route noise impact through heatmaps, trajectory visualization, and per-timestep perturbation measurements.

Our DeepStackEnsemble achieves MCC 0.859, a 13.6% relative improvement over classical baselines. The remainder of this paper covers related work (Section 2), dataset and methodology (Section 3), experimental results (Section 4), and conclusions (Section 5).

II. RELATED WORK

Passenger counting spans multiple sensing modalities. Video-based APC using YOLOv5 achieves high accuracy but degrades under occlusion and variable lighting [2]. Edge AI cameras address scalability [3]. ML pipelines on traditional APC data enable offline correction of sensor errors [4].

Wi-Fi sensing avoids the cost and privacy drawbacks of cameras. iABACUS [5] demonstrated counting via Wi-Fi probe requests. Myrvoll et al. [6] used Wi-Fi signature counts for ridership estimation. Fabre et al. [7] applied random forests and gradient boosting to Wi-Fi sensor data. Simonic et al. [8] monitored presence through passive probe requests. CSI-based methods [9] offer finer signal characterization but require specialized hardware unavailable on commodity access points. RSSI fingerprinting for localization has been extensively surveyed [10].

Recent advances in sequence modeling inform our architectural choices. Mamba [11] introduced selective state-space models with $O(L)$ complexity. Mamba-3 MIMO [12] extended this paradigm with multi-input multi-output capabilities. Autoformer [13] replaced self-attention with FFT-based auto-correlation mechanisms. Mixture of Experts [14] enables conditional computation through sparse activation.

Our prior work [1] established classical ML baselines on this dataset. Across 34 algorithms, Gaussian Processes achieved the best combined MCC of 0.756. This paper extends

that analysis with 11+ specialized DL architectures, comprehensive HPO via Optuna [15], and systematic interference analysis.

III. MATERIALS AND METHODS

A. Dataset and Feature Engineering

The dataset simulates public transport in a controlled environment [1]. Zone A represents the vehicle interior with an access point near the door. Zone B represents the bus stop in an adjacent corridor. Four devices execute four movement patterns, denoted AA (inside vehicle), BB (at stop), BA (boarding), and AB (alighting). Each device produces 10 RSSI readings at 1 s intervals per observation.

Two scenarios were collected. The pure scenario uses a single device with no interference (160 samples). The noise scenario uses four concurrent devices executing different route combinations (1,200 samples). The combined dataset contains 2,253 labeled samples distributed as AA 407, AB 387, BA 399, and BB 417.

Four engineered channels are extracted from the raw RSSI sequence r_t . The raw RSSI r_t provides absolute signal position. Velocity $\Delta_t = r_t - r_{t-1}$ captures movement speed. Acceleration Δ_t^2 captures rate of change. Window deviation $d_t = r_t - \mu_w$, where μ_w is the per-sample mean, captures trajectory shape independent of absolute RSSI level. These channels yield input tensors of shape $(N, 10, 4)$.

B. Proposed Architectures

Table I lists the 11+ evaluated architectures. All models output 4-class logits through a linear classification head.

Table I
DL ARCHITECTURES EVALUATED

Architecture	Type	Key characteristic
RNN, GRU, LSTM, BiLSTM	Recurrent	2 layers, hidden 64
TCN [16]	Convolutional	Dilated causal, residual
Transformer	Attention	4-head self-attention, sinusoidal PE
MambaClassifier [12]	State-space	$O(L)$ complexity, rank- $R=16$
CNN2DRNN	Hybrid 2D+RNN	Conv2d (3,3),(5,3) $\rightarrow$ LSTM/GRU + attention
MetaFusion	Context-aware	LSTM/GRU + noise/path embeddings (dim 8)
Multi-Scale CNN	Multi-branch	Kernels (3,5,7) + residual + gating
SoftMoE	Mixture of Experts	Heterogeneous experts, top-2 gating
Autoformer [13]	Decomposition	FFT auto-correlation, trend-seasonal blocks
DeepStackEnsemble	Two-stage meta	18 bases $\rightarrow$ per-class Level2Net $\rightarrow$ residual MLP

The DeepStackEnsemble uses per-class Level2Nets intentionally. Confusion pairs such as AA $\leftrightarrow$ BA (both near-AP) require different base model weighting than AB $\leftrightarrow$ BB (both exterior). The residual skip connection preserves calibrated per-class logits while the MetaMLP corrects systematic biases.

C. Augmentation and Implementation

Two strategies expand the training data from 2,253 to 10,115 samples. Class-conditioned Gaussian noise adds 6 copies to clean samples ($\sigma = 3.0$ dBm) and 4 copies to noisy samples ($\sigma = 0.8$ dBm). Intra-class Mixup blends 600 random pairs per class as $X_{\text{new}} = \lambda X_1 + (1 - \lambda) X_2$ with $\lambda \sim \text{Beta}(0.4, 0.4)$, restricted to same-label pairs.

The pipeline uses PyTorch 2.6 and Optuna 4.7. Training uses AdamW optimizer with cosine annealing, early stopping (patience 25), label smoothing ($\epsilon = 0.1$), Mixup augmentation ($\alpha = 0.2$), L1 regularization ($\lambda = 10^{-5}$), L2 weight decay (10^{-4}), dropout ($p = 0.3$), gradient clipping at 1.0, and batch size 64. HPO uses 50 Optuna TPE trials per model. SoftMoE additionally uses NSGA-II multi-objective optimization. The

primary metric is MCC [17]. All experiments use 5-fold stratified cross-validation with 80/20 splits and 3 random seeds (3, 5, 42). A total of 65 models were evaluated.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

Table II describes the four dataset variants. Each stresses a different aspect of model robustness, from clean single-device recording to dense multi-device interference and synthetic augmentation. All models use 5-fold stratified cross-validation with 80/20 splits. Results are reported at seed 42 unless stated otherwise.

Table II
DATASET VARIANTS

Variant	Train	Test	Condition
Pure	128	32	Single device, no interference
Noise	960	240	Four concurrent devices
Normal	1,802	451	Original unbalanced combined
Augmented	8,090	2,023	Synthetic samples added to training

B. Variant-Level Comparison

Figure 1 shows the best model MCC achieved on each variant. Pure conditions yield RNN at MCC 0.881 (seed 3). Multi-device noise degrades performance to DS RNN_B at 0.798, a 9.4% absolute decline. The normal variant reaches DeepStackEnsemble at 0.859. Data augmentation lifts HPO TCN to 0.865.

Table III compares these results against classical ML baselines from our prior work [1]. DL improves combined data by 13.6% and noise by 3.6%, but trails on pure data by 4.4%. DL's advantage emerges where both training data and interference complexity are sufficient; on clean, data-scarce problems, simpler methods prevail.

Table III
DL VS. CLASSICAL ML (MCC)

Variant	Classical best	DL best	Change
Combined	0.756 (Gaussian Process)	0.859 (DeepStack)	+13.6%
Pure	0.922 (KNN)	0.881 (RNN)	-\$4.4%
Noise	0.770 (CatBoost)	0.798 (DS RNN_B)	+3.6%

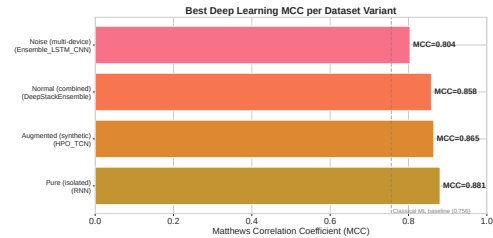


Figure 1. Best model MCC per variant.

C. Architecture Performance and HPO

Table IV lists the top models per variant. Among single architectures, CNN (MCC 0.838) and TCN (0.836) lead. Mamba reaches 0.800, outperforming the Transformer at 0.713. DeepStackEnsemble achieves the best combined MCC of 0.859.

Table IV
TOP MODELS BY VARIANT

Model	Variant	MCC	Acc.
DeepStackEnsemble	Normal	0.8585	0.8937
HPO GRU	Normal	0.8524	0.8893
HPO TCN	Normal	0.8470	0.8843
CNN	Normal	0.8383	0.8774
HPO TCN	Augmented	0.8647	0.8982
HPO CNN	Augmented	0.8642	0.8977
Ensemble CNN+TCN	Augmented	0.8587	0.8932
RNN	Pure*	0.8808	0.9063
DS RNN_B	Noise*	0.7981	0.8458
HPO LSTM	Noise*	0.7908	0.8417

Table V quantifies the impact of hyperparameter optimization. Recurrent architectures benefit most. GRU improves from 0.710 to 0.852 (20.0% gain). LSTM improves from 0.686 to 0.844 (23.1% gain). BiLSTM improves from 0.684 to 0.820 (19.9% gain). These gains reveal severe underparameterisation at default settings. CNN and TCN gain less because their inductive biases are nearer to optimal out of the box. Each model underwent 50 Optuna TPE trials.

Table V
HPO GAINS ON NORMAL VARIANT

Model	Default MCC	HPO MCC	Gain
GRU	0.7101	0.8524	+20.0%
LSTM	0.6855	0.8435	+23.1%
BiLSTM	0.6838	0.8195	+19.9%
RNN	0.7552	0.8319	+10.2%
Mamba	0.8002	0.8418	+5.2%
TCN	0.8356	0.8470	+1.4%
CNN	0.8383	0.8357	−0.3%

D. Augmentation Ablation

Table VI shows augmentation impact. CNNs and TCNs gain from synthetic diversity (0.011 to 0.029 MCC), while recurrents may degrade slightly (0.001 to 0.006 MCC). Synthetic noise creates local fluctuations that recurrents interpret as temporal dependencies; CNNs smooth them through weight sharing.

Table VI
AUGMENTATION IMPACT (NORMAL VS. AUGMENTED MCC)

Model	Normal	Augmented	Change
CNN	0.8383	0.8490	+0.011
TCN	0.8356	0.8344	−0.001
GRU	0.7101	0.7039	−0.006
LSTM	0.6855	0.6804	−0.005
Mamba	0.8002	0.8073	+0.007
HPO TCN	0.8470	0.8647	+0.018
HPO CNN	0.8357	0.8642	+0.029

E. Ensemble Strategies

Table VII compares ensemble strategies. The progression from CNN (0.838) through HPO GRU (0.852) and stacking (0.849) to DeepStack (0.859) shows incremental gains. Uniform voting across all nine architectures degrades to 0.807, below a single CNN, because weak classifiers (Transformer 0.713, BiLSTM 0.684) contribute noisy probabilities. DeepStack avoids this through per-class learned weighting.

Table VII
ENSEMBLE STRATEGY COMPARISON (NORMAL VARIANT)

Strategy	MCC	Description
CNN (best single)	0.8383	No HPO or ensemble
HPO GRU (best HPO)	0.8524	50 Optuna trials
Stacking (LR)	0.8488	Logistic regression on 9 logits
DeepStackEnsemble	0.8585	Two-stage learned weighting
Ensemble All-9 (vote)	0.8074	Uniform soft voting

F. Per-Class Performance

Table VIII presents per-class F1 scores. AB (alighting) leads at 0.893 to 0.925 due to its distinctive descending RSSI trajectory. AA (inside vehicle) is hardest at 0.831 to 0.894. Sustained high RSSI near the access point saturates the signal, causing confusion with BA. Both classes involve near-AP states. Noise degrades AB most because concurrent near-AP transmissions mask its characteristic 20 dBm descent.

Table VIII
PER-CLASS F1 BY VARIANT (BEST MODEL)

Variant	AA	AB	BA	BB
Pure	0.894	0.925	0.914	0.907
Noise	0.831	0.893	0.838	0.818
Normal	0.866	0.925	0.895	0.873
Augmented	0.866	0.925	0.895	0.873

G. Noise and Interference Analysis

Figure 2 quantifies the mean RSSI shift induced by each concurrent interferer on each primary route. Three patterns stand out. First, AA dominates as interferer, producing shifts of 45 to 53 dBm across all routes. Devices inside the vehicle transmit at high power near the access point. BB causes minimal shift, at most 3.1 dBm, because devices at the stop are farthest from the AP. Second, AB is most affected. AA interference adds 46.6 dBm, masking the characteristic descent. Third, self-interference is minimal. Diagonal entries, where interferer and primary route are the same class executed by different devices, show shifts of only 1.2 to 4.6 dBm.

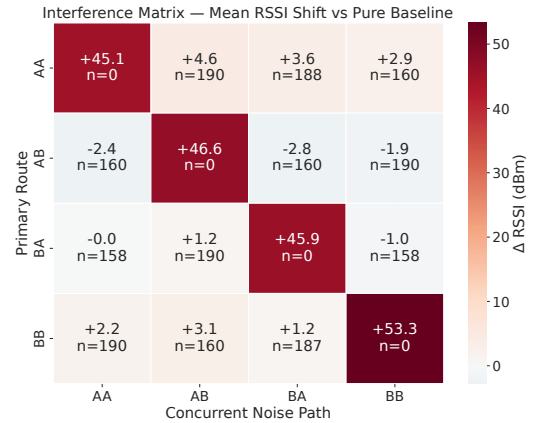


Figure 2. Interference heatmap: mean RSSI shift per (Route × Interferer) pair.

Figure 3 shows interference effects per class. AA: BA interference raises the high-RSSI profile, blurring near-AP states. AB: AA flattens the 20 dBm descent into a constant profile, directly causing AB’s noise-induced F1 drop. BA: AA compresses the ascent. BB: the flat pattern shifts but retains

shape; classification depends on temporal flatness, not absolute level, explaining BB’s noise resilience.

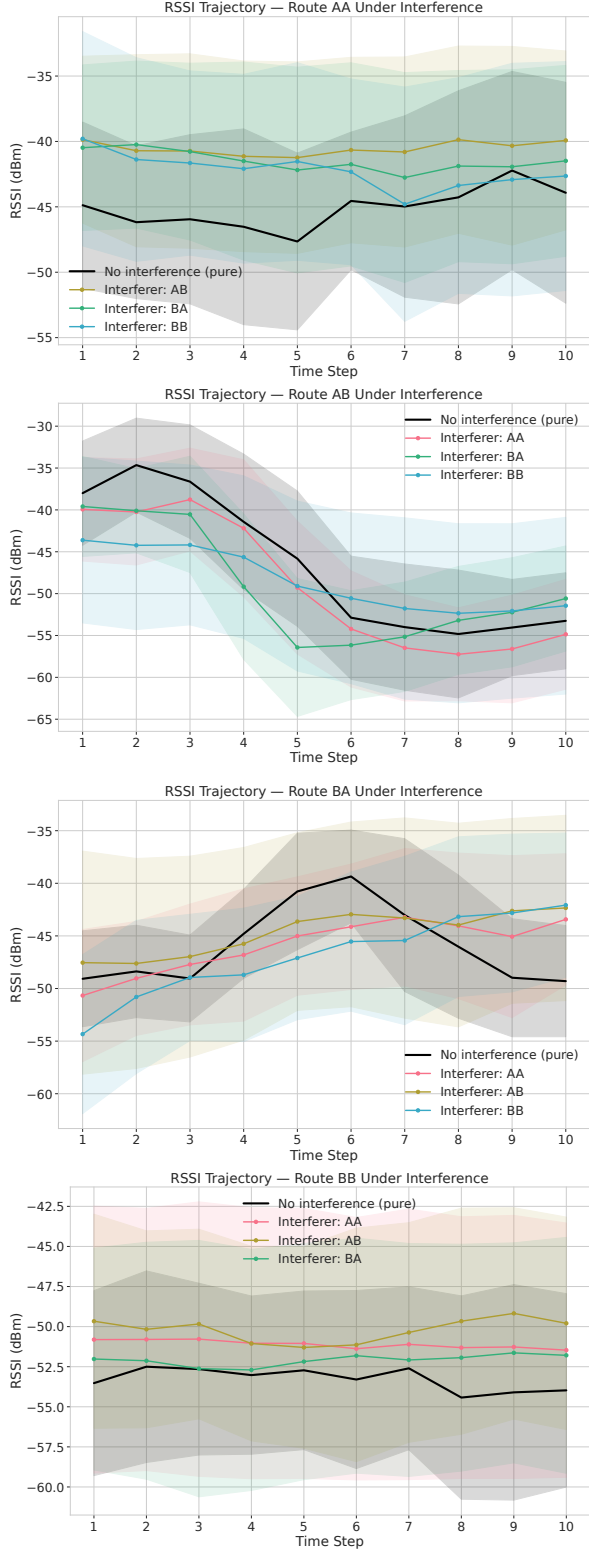


Figure 3. Trajectories under interference for all four classes.

Per-timestep perturbation peaks at timesteps 4 to 6, when devices are closest to the access point. MetaFusion_GRU reaches MCC 0.783 on the noise variant, compared to 0.758

for standard GRU. Encoding noise context as learned embeddings provides partial mitigation.

H. Feature Importance

Figure 4 shows permutation importance across the four engineered channels and ten timesteps, averaged over CNN, TCN, and GRU. Window deviation dominates early timesteps ($t = 1$ to 3), capturing each class’s characteristic offset from its own baseline. Velocity peaks at $t = 4$, the mid-trajectory moment of maximum movement. Raw RSSI matters most at $t = 6$, where absolute levels separate near-AP positions from exterior ones. Acceleration contributes at $t = 3$ to 5.

These results extend prior findings [1], where classical ML on raw RSSI alone identified early timesteps as most important (MCC 0.756). DL additionally captures mid-trajectory dynamics via the velocity channel (at $t = 4$, $\Delta\text{MCC } 18.4 \times 10^{-3}$) that classical models, lacking derivative features, could not exploit. The window deviation channel explains BB’s noise resilience: trajectory shape survives interference-level shifts that distort absolute RSSI.

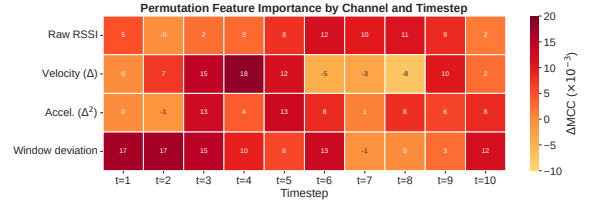


Figure 4. Permutation feature importance by channel and timestep (CNN, TCN, GRU average).

I. Discussion

Classical and deep methods complement each other. KNN leads on clean single-device data (0.922 vs. 0.881) because local neighbourhood structure directly captures clean RSSI trajectories without training. DeepStack leads under multi-device interference (0.859 vs. 0.770) because learned feature hierarchies disentangle overlapping signals. This complementarity suggests density-based routing: KNN for sparse single-device conditions, DeepStack for dense multi-device scenarios.

The HPO gain gap between recurrent architectures (20–23%) and CNN/TCN ($\pm 1\%$) reflects a fundamental difference in inductive bias. CNN kernels operate on fixed-size windows regardless of sequence length. TCN dilated convolutions naturally match 10-step sequences through exponentially growing receptive fields. Recurrent hidden states must encode all temporal context into a fixed-size vector. At default dimension 64, this creates an information bottleneck. HPO discovers larger hidden dimensions that remove the bottleneck. The augmentation results reinforce this pattern: models with strong inductive biases extract useful signal from synthetic diversity (HPO CNN +0.029, HPO TCN +0.018); recurrences without HPO are confused by it (GRU −0.006).

Uniform voting across all nine architectures (MCC 0.807) underperforms a single CNN (0.838). Weak classifiers contribute noisy probability estimates that shift the decision boundary. DeepStack (0.859) avoids this through per-class Level2Nets that learn class-specific base model importance. Two findings point toward future work. Mamba’s $O(L)$ complexity enables scaling to longer observation windows of 20–30 seconds, where self-attention’s quadratic cost becomes

prohibitive. MetaFusion_GRU (MCC 0.783 on noise vs. 0.758 baseline) demonstrates that context-aware architectures incorporating noise metadata can partially disentangle interference.

Limitations include the controlled laboratory environment, fixed 10-second window, MAC address randomisation, and untested device heterogeneity across smartphone models. For deployment, CNN (28K parameters) enables real-time on-device inference at the access point. DeepStackEnsemble (1.2M parameters) targets server-side batch inference where latency permits evaluating all 18 base models.

V. CONCLUSION

This paper evaluated deep learning architectures for RSSI-based passenger movement classification in public transport. Building on classical ML baselines at MCC 0.756, we tested 11+ architectures spanning recurrent, convolutional, transformer, state-space, and mixture-of-experts models. The evaluation used four dataset variants, class-conditioned data augmentation, and exhaustive hyperparameter optimization.

The two-stage DeepStackEnsemble achieved MCC 0.859 and accuracy 89.4% on combined data, a 13.6% relative improvement. State-space models reached up to MCC 0.850 with $O(L)$ complexity, confirming their viability as an alternative to transformers for resource-constrained deployment.

Cross-route interference analysis revealed a 9.4% absolute MCC degradation under multi-device noise. Heatmap analysis identified AA as the dominant interferer and AB as the most vulnerable route. Trajectory visualization showed that concurrent near-AP transmissions erase AB's characteristic 20 dBm RSSI descent. BB proved most noise-resilient because its classification signal comes from temporal flatness, which survives interference-level shifts intact.

Feature importance analysis confirmed and extended prior findings. Classical ML identified early timesteps as most informative. DL additionally captured mid-trajectory motion dynamics through the velocity channel, contributing directly to the 13.6% improvement. Window deviation emerged as the most informative channel, explaining BB's noise resilience through shape-based classification.

Future work should explore attention mechanisms that learn to down-weight interfered timesteps. Multi-AP fusion could disambiguate near-AP states through spatial diversity. Integration of movement classification with origin-destination matrix inference would enable end-to-end passenger flow modeling. Context-aware architectures that jointly model signal and environmental metadata offer a promising direction for real-world deployment.

VI. GENERATIVE AI DECLARATION

During the preparation of this manuscript, the authors used DeepSeek as a writing assistant for language improvement, grammar correction, and formatting. All technical content, experimental design, model architectures, results, analysis, and conclusions represent the authors' original intellectual contribution. The authors take full responsibility for the accuracy and integrity of the content. No AI-generated figures, fabricated citations, or manipulated results are present in this work.

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